Subsonic Low Reynolds Number: E387 Airfoil

Case Details

The objective of the simulation is to validate the E387 airfoil's experimental data given in [1] with low computational cost for fast but accurate results.

- Reference: `https://m-selig.ae.illinois.edu/uiuc_lsat/Low-Speed-Airfoil-Data-V1.pdf`
- Airfoil: E387 airfoil
- Mach Number: 0.15
- Reynolds Number: $60,000$
- Angle of Attack: $0°$

How to Run

1. Select E387 airfoil from the airfoil directory
2. Enter the desired number of coordinates if required and output directory.
3. Click on "Analyze" to obtain the accurate parameterization method and its respective parameters
4. Go to SU2 Settings → Meshing & Conditions
5. Enter Mach, Reynolds number and enter 0 in minimum AoA with 0 step.
6. Select "Load recommended settings"
7. Choose the type of mesh and target y+ value
8. Select "Generate Structured Mesh"
9. Go to SU2 Settings → Solver Setup & Run
10. Modify settings as desired or select "Run SU2 Analysis" (validated settings have been provided as default and are given here in the next section)
11. After convergence, the output or post-processed contours will be displayed in the SU2 Settings → Results tab and saved in the output directory.

Solver Settings

Given below are the solver settings validated for this case and are provided by default for this regime. This case is run using unsteady parameters:

Setting	**File (LowRe.cfg**
NUM_METHOD_GRAD	WEIGHTED_LEAST_SQUARES
CONV_NUM_METHOD_FLOW	L2ROE
SLOPE_LIMITER_FLOW	VENKATAKRISHNAN
LINEAR_SOLVER	FGMRES
LINEAR_SOLVER_PREC	ILU
KIND_TURB_MODEL	SST
KIND_TRANS_MODEL	LM
CFL_NUMBER	1 (Adpative)
CONV_FIELD	RMS_DENSITY
Min log10 Residual	-8
TIME STEPPING	YES

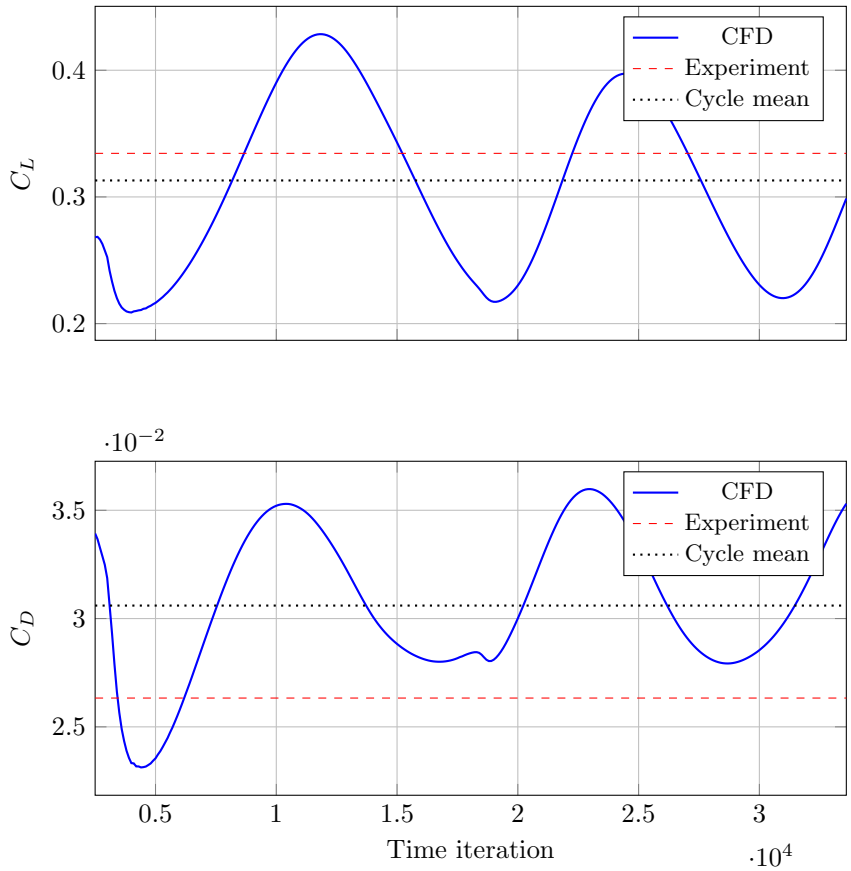
Table 1: Solver settings used

Result

AoA	Cl_{exp}	Cd_{exp}	Cl_{obt}	Cd_{obt}	**Cl Error (%)**	**Cd Error (%)**
0	0.3344	0.0306	0.313	0.02633	6.4	16

Table 2: Results at Mach 0.15 with comparison between experimental and obtained data

The high error in Cd is a known result in this regime. The laminar speration bubble grows in size and sheds. The bubble reduces the camber of the airfoil, increasing drag and reducing lift. The prediction in reattachment is challenging in this Reynolds number [2]. Depending on the bubble getting re-attached or not, the magnitude varies, leading to discrepancies in not just simulations but experiments also [3]. The transition model (Langtry-Menter) is necessary in this case for accurate prediction of the bubble, however it requires tuning using the turbulence intensity parameter (0.3%) used here [4]. This tuning requires a separate study and is not included here, however, the platform can be used to test these cases conveniently.



The separation point (x/c) was found to reduce with iterations, with $x/c = 0.45$ from 0.47, towards the end of two cycles shown in Figure. Cl and Cd values are found to oscillate about the reported mean in but its physical vs numerical nature is a subject of further investigation, in conjunction with mesh studies.

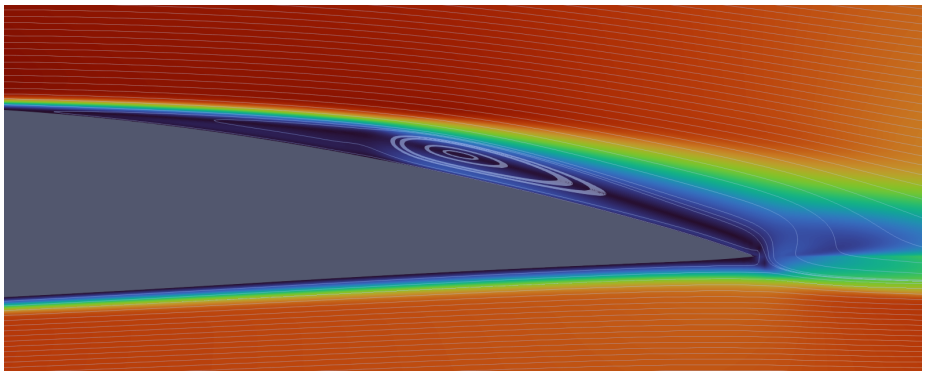


Figure 2: Laminar separation bubble visualized using streamlines

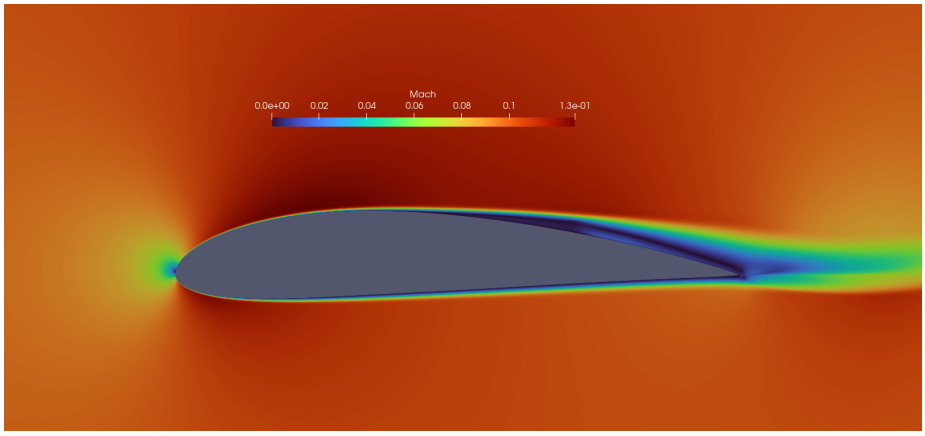


Figure 1: Velocity field as visualized in ParaView

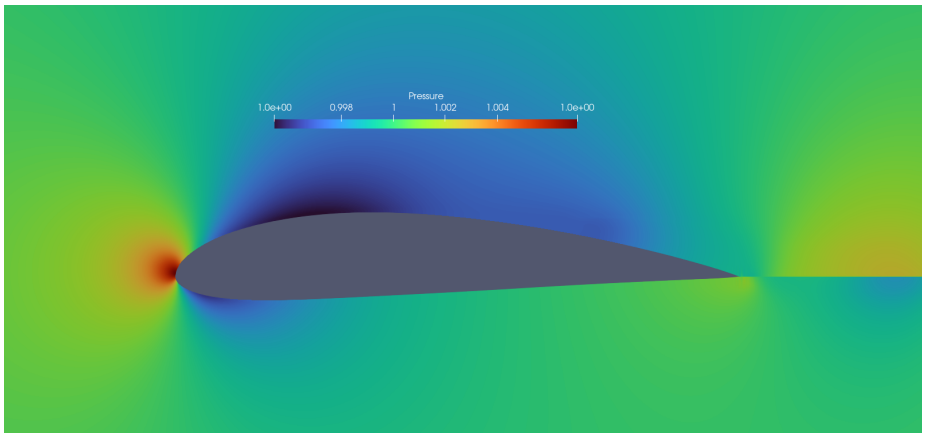


Figure 3: Pressure field as visualized in ParaView

References

[1] M. S. Selig, *UIUC Airfoil Data Site*, Department of Aerospace Engineering, University of Illinois at Urbana-Champaign, 1996. Available online at `https://m-selig.ae.illinois.edu/ads.html`.

[2] E. A. Sezer, M. Ayvazoglu, M. Hatem, S. Keskin, M. Ozden, and M. S. Genç, "Modeling of the Separation Bubble on Cambered Airfoils Utilizing Modified Parameters in a Transition Model," *CMES-Computer Modeling in Engineering & Sciences*, vol. 146, no. 3, pp. 66796, 2026, doi: 10.32604/cmes.2026.076446.

[3] R. J. McGhee, B. S. Walker, and B. F. Millard, "Experimental Results for the Eppler 387 Airfoil at Low Reynolds Numbers in the Langley Low-Turbulence Pressure Tunnel," NASA Langley Research Center, Hampton, VA, Tech. Rep. NASA-TM-4062, Oct. 1988.

[4] M. Anıl and S. Sezer, "Modeling of the Separation Bubble on Cambered Airfoils Utilizing Modified Parameters in a Transition Model," *CMES - Computer Modeling in Engineering and Sciences*, vol. 146, no. 3, 2026. `https://doi.org`